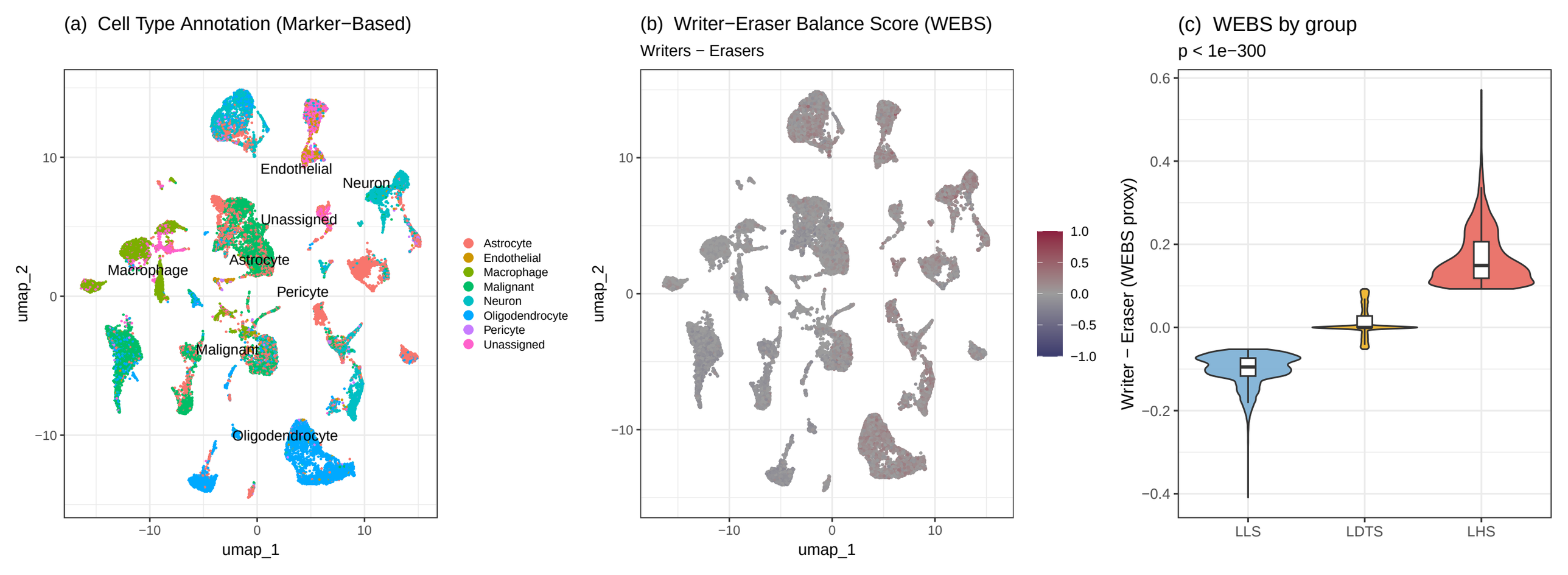

(a) Cell Type Annotation (Marker−Based)
Endothelial
Neuron
Unassigned
Astrocyte
Macrophage
Pericyte
Malignant
Oligodendrocyte
umap_1
umap_2
Astrocyte
Endothelial
Macrophage
Malignant
Neuron
Oligodendrocyte
Pericyte
Unassigned
(b) Writer−Eraser Balance Score (WEBS)
Writers − Erasers
1.0
0.5
0.0
−0.5
−1.0
(c) WEBS by group
p < 1e−300
Writer − Eraser (WEBS proxy)
LLS
LDTS
LHS